AI INFRASTRUCTURE INTELLIGENCE

HydroGuard AI

PROBLEM & SOLUTION

The Cooling Challenge

- AI data centers depend on liquid cooling to prevent GPU overheating.
- Hidden pump failures (micro-cavitation) evade standard thermal monitoring.
- Sudden failures trigger downtime and SLA losses of up to $30,000.

HydroGuard AI Solution

HydroGuard AI detects early cooling system failures via ultrasonic telemetry and automatically transfers workloads from failing racks to healthy ones.

Automated workload migration for zero-downtime continuity

DIAGNOSTIC ARCHITECTURE

Diagnostics

High-precision real-time analysis for maximum infrastructure protection.

PILLAR 01

Detect

Identify abnormal ultrasonic frequencies (42.5 kHz) before GPU temperatures spike.

Real-time Telemetry

PILLAR 02

Categorize

Classify fault type: Micro-Cavitation, Fluid Viscosity Loss, or Bearing Friction.

Fault Classification

PILLAR 03

Prioritize

Flag HIGH RISK on the failing rack (paid AI workload). Prevents $30,000 potential SLA loss.

Critical SLA Alert

PILLAR 04

Explain

AI failure explanation & action recommendation to migrate Rack G-04 to Rack G-08.

Auto Workload Transfer

LIVE SYSTEM SHOWCASE

Demonstration